MOTS-c Mitochondrial Peptide

Step-by-Step Reconstitution Process

- **Gather supplies: BAC water, alcohol swabs, insulin syringe, vial**
- Removing plastic cap. Carefully pop off the plastic cap, do not remove metal flange.
- **Clean both vials:** Wipe the rubber stoppers with alcohol swabs
- **Draw BAC water:** Pull your chosen amount into the syringe
- **Inject slowly:** Add the water to the peptide vial, aiming it at the glass wall—not directly onto the powder
- **Swirl gently:** Roll the vial between your palms. Never shake.
- **Wait:** Let it sit 5-10 minutes. Solution should be crystal clear.
- **Store:** Refrigerate immediately. Good for 28-30 days once reconstituted.

If it's cloudy or won't dissolve: Let it sit longer. Try gentle warming (under your arm or in lukewarm water). If still cloudy after 30 minutes, contact your source.

Practical Protocols

The following protocols are not derived from published dose-finding studies. They represent clinical practice patterns and community consensus.

STANDARD PROTOCOL

Dose: 5 to 10 mg per week total
Frequency: Split into 3 injections per week (Monday, Wednesday, Friday)
Timing: Morning on an empty stomach, or 30 minutes before training
Administration: Subcutaneous injection
Duration: 4 to 8 week cycles followed by equal time off

BY GOAL

Training enhancement (active exerciser): 5 to 10 mg per week, 3 injections
Metabolic support (insulin resistance): 10 to 15 mg per week, 3 injections
Longevity and general wellness: 5 to 10 mg per week, 3 injections, 4 to 8 week cycles

RECONSTITUTION

Vile size	10mg	
BAC	1ml	1mg = 10 units
		2mg = 20 units
		5mg = 50 units

Common Questions

Why did MOTS-c not work for me? Three main reasons. First, if your mitochondria are damaged from years of poor habits, MOTS-c has no functional machinery to optimize. Address that first with SS-31. Second, if you are not exercising, your AMPK pathway may be suppressed, which means the signal has more resistance. Exercise is non-negotiable. Third, if you are dosing once a week, your circulating levels are only active for a few hours out of 168. Split your dose to at least three times per week.

Do I need to take SS-31 before MOTS-c? It depends on how much damage you are working with. If you have significant mitochondrial dysfunction from years of poor habits, chronic fatigue, or metabolic disease, then yes, sequencing SS-31 first for 4 to 8 weeks to repair the structural damage before adding MOTS-c is still the right call. But if you have moderate age-related decline and are otherwise active and functional, newer research shows that MOTS-c and SS-31 reduce ROS through completely non-overlapping mechanisms, which means running them concurrently is a viable option. SS-31 repairs the membrane, MOTS-c improves functional efficiency and upregulates antioxidant defenses at the nuclear level. Two angles on the same problem at the same time. If you are younger and metabolically healthy, you can start with MOTS-c directly.

Is MOTS-c an exercise replacement? No. It is an exercise mimetic, meaning it copies some of what exercise does at the cellular level. But exercise produces a 12-fold increase in endogenous MOTS-c production along with hundreds of other beneficial signals. MOTS-c enhances what exercise does. It does not replace it. People who exercise regularly get better results from MOTS-c than sedentary users.

Can I take MOTS-c long term? MOTS-c can be taken for longer cycles than many other peptides, but periodic breaks are recommended. A typical approach is 4 to 8 weeks on, then equal time off. Some people run it for 12 to 16 weeks with breaks. The concern with indefinite use is potential tolerance and the fact that long-term effects are unknown in humans.

When should I take MOTS-c? Morning on an empty stomach is typical. This mimics the natural increase that occurs during fasting. Many people report better results when dosing 30 minutes before exercise, particularly fasted cardio. Avoid late-day dosing as some people report it can disrupt sleep.

Does MOTS-c affect blood sugar? Yes. MOTS-c improves insulin sensitivity and increases glucose uptake in muscle cells. If you take diabetes medications, monitor your blood sugar closely. The combination could cause hypoglycemia if your medication dose is not adjusted.

Does MOTS-c cause cancer? No study has demonstrated that MOTS-c promotes cancer. In fact, the available evidence trends in the opposite direction. A 2024 study found MOTS-c suppressed ovarian cancer progression, and low MOTS-c levels correlated with poor prognosis in cancer patients. MOTS-c activates AMPK, which is generally considered tumor-suppressive. However, the research is still limited and the long-term effects are unknown. If you have active cancer or a recent history of malignancy, avoid this compound until cleared by your oncologist.

Side Effects

What Users Report

Common: Injection site reactions (redness, swelling, mild bruising), mild fatigue or lethargy when first starting (your body adjusting to AMPK activation), changes in appetite (some report suppression, others report increased hunger), mild headaches
Less common: Heart palpitations, sleep disturbance if dosed late in the day, mild GI discomfort (nausea, bloating), muscle cramping, dizziness
Most side effects appear to be temporary, typically resolving within the first few days as your body adjusts.